



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

BDCTD301

CONSTRUCTION BASIC TECHNICAL DRAWING

REALIZE CONSTRUCTION BASIC TECHNICAL DRAWING

Competence

RQF Level: 3

Learning Hours



Credits: 9

Sector: Construction and Building Services

Trade: Building construction

Module Type: Specific

Curriculum: CBSBDC3001- TVET Certificate 3 in Building Construction

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Purpose statement	This core module describes the skills, knowledge and attitude to be acquired by the learner to draw and interpret construction drawings at construction site or in the office in order to perform the construction work efficiently.					
Leaning assumed to be in place	N/A					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	100%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment			N/A

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria		
1. Identify materials, instrument and Equipment.	1.1.	Proper identification of various drawing instruments	
	1.2.	Proper identification of various drawing equipment	
	1.3.	Proper identification of drawing materials according to the task provided	
2. Use construction drawing, materials, instruments and equipment.	2.1.	Correct use of drawing materials, instruments and equipment	
	2.2.	Correct maintenance of Instruments and equipment	
	2.3.	Efficient checking the accuracy of instruments and equipment	
3. Interpret lines and symbols	3.1.	Types of lines are interpreted properly according to the use	
	3.2.	Sizes of line are precisely interpreted according to the use	
	3.3.	Drawing symbols used in construction drawing are accurately interpreted with respect to their use	

4. Apply drawing scales, dimensions and lettering	4.1. Scale is accurately applied in construction drawing
	4.2. Dimension is systematically applied to construction drawings
	4.3. Lettering is conveniently done according to drawing standards
5. Perform angle projection	5.1. Angle types are properly differentiated according to the standard
	5.2. Neat geometrical figures and objects are precisely drawn as per rules
	5.3. Make proper projection according to their types
6. Perform construction drawing plans	6.1. Drawing sheets layout is accurately prepared with respect to the drawing
	6.2. Construction drawing plans are precisely identified with respect to the drawing
	6.3. Neat construction drawing plans are carefully drawn as per their use
	6.4. Interpretation of construction drawing plans is correctly done according to the titles block

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Identify materials, instruments and equipment 2. Use construction drawing, materials, instruments and equipment. 3. Interpret lines and symbols Construct kiln 4. Apply drawing scales, dimensions and lettering 5. Perform angle projection 6. Perform construction drawing plans
Learning outcome 1: Identify materials, instrument and equipment	Learning hours: 5
Indicative content	
• Identification of drawing instrument -Writing instruments (ink pen, pencils...) -Tracing instruments (Rulers, set square...) -Accessory instruments (Template, knife...) • Identification of drawing Equipment -Drawing Board • Identification of drawing materials -Types of drawing papers -Drawing papers format and sizes	
Resources required for the learning outcome	
Equipment	Drawing pins and clips, Compass, Drawing pencil , Drawing board, Drawing tables
Materials	Construction drawing handout, A3 Papers, A2 Papers, A1 Papers, A0 Papers, Inks, Eraser, Pencil lead, Lead
Tools	Books, Internet, Handouts, Ink-pen, Pencils, set squares, Protractor, Compass, French curves, Template, Pantograph, Proportion divider, Scale rule, Pencil, sharpener, making tape, Drawing board, Knife, Scissors
`	• Group discussion • Trainer guided • Matching questions • Multiple choice • Sentence completion • Short questions

	• Essay • Open ended questions
Formative assessment methods	• Written assessment • Performance assessment

Learning outcome 2: Use of drawing materials, instruments and equipment		Learning hours: 5
Indicative content		
• Function of drawing instruments, materials and equipment • Maintenance of drawing equipment and instruments • Examination of status of drawing tools, instruments and equipment • Calibration of drawing instrument and equipment • Re- calibration of status of drawing tools, instruments and equipment		
Resources required for the indicative content		
Equipment	Drawing pins and clips, Compass, Drawing pencil , Drawing board, Drawing tables	
Materials	Construction drawing handout, A3 Papers, A2 Papers, A1 Papers, A0 Papers, Inks, Eraser, Pencil lead, Lead	
Tools	Books, Internet, Handouts, Ink-pen, Pencils, set squares, Protractor, Compass, French curves, Template, Pantograph, Proportion divider, Scale rule, Pencil, sharpener, making tape, Drawing board, Knife, Scissors	
Facilitation techniques	• Group discussion • Trainer guided • Matching questions • Multiple choice • Sentence completion • Short questions • Essay • Open ended questions	
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment	

Learning outcome 3: Interpret lines and symbols	Learning hours: 5
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Indicative content	
• Description of drawing lines ➤ Definition of lines used in drawing ➤ Types of lines used in drawing ➤ Use of drawing lines types • Identification of size of lines • Definition of symbols for construction drawing • Functions of symbols for construction drawing • Types of symbols used in construction drawing	
Resources required for the indicative content	
Equipment	Drawing pins and clips, Compass, Drawing pencil , Drawing board, Drawing tables
Materials	Construction drawing handout, A3 Papers, A2 Papers, A1 Papers, A0 Papers, Inks, Eraser, Pencil lead, Lead
Tools	Books, Internet, Handouts, Ink-pen, Pencils, set squares, Protractor, Compass, French curves, Template, Pantograph, Proportion divider, Scale rule, Pencil, sharpener, making tape, Drawing board, Knife, Scissors
Facilitation techniques	• Lectures • Demonstration and simulation • Individual and group work • Practical exercise • Individualized • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment • Project based assessment
Learning outcome 4: Apply drawing scales, dimensions and lettering	Learning hours: 15
Indicative content	
• Description of drawing scales ➤ Definition of scale ➤ Function of scale • Sizes of scales	

- Full size scales
- Reducing scales
- Enlarging scales
- **Scale representation**
 - Fractional Representation
 - Ratio representation
 - Graphical representation
- **Drawing paper presentation**
 - Drawing paper layout and margin
 - Drawing sheet frame
 - Title block
 - Element of the title block
 - Folding drawing sheet
- **Types of dimensions**
 - Size dimensions
 - Location dimensions
- **Method of dimensioning**
 - Aligned method
 - Unidirectional method
- **Elements of dimensioning**
 - Dimension line
 - Leader
 - Extension line
- **Function Features of lettering**
 - Uniformity
 - Neatness
 - Rapidity
- **Classification of lettering**
 - Gothic lettering
 - Roman lettering
- **Style of freehand lettering**
 - Vertical or upright freehand lettering
 - Inclined or italic freehand lettering

Resources required for the indicative content

Equipment	Drawing pins and clips, Compass, Drawing pencil , Drawing board, Drawing tables
Materials	Construction drawing handout, A3 Papers, A2 Papers, A1 Papers, A0 Papers, Inks, Eraser, Pencil lead, Lead
Tools	Books, Internet, Handouts, Ink-pen, Pencils, set squares, Protractor, Compass,

	French curves, Template, Pantograph, Proportion divider, Scale rule, Pencil, sharpener, making tape, Drawing board, Knife, Scissors
Facilitation techniques	• Lectures • Demonstration and simulation • Individual and group work • Practical exercise • Individualized • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment • Project based assessment

Learning outcome 5: Perform angle projection		Learning hours: 20
Indicative content		
• Dimensions of object views ➤ 2d dimension, ➤ 3d dimension • Plane of projections ➤ Vertical plane ➤ Horizontal plane • Angles of projections ➤ First angle projection ➤ Third angle projection • Drawing of lines • Drawing of triangle • Drawing of a circle • Polygons • Conics sections • Solids • Types of projections • Views and sections of objects • Application of projections		
Resources required for the indicative content		
Equipment	Drawing pins and clips, Compass, Drawing pencil , Drawing board, Drawing tables	
Materials	Construction drawing handout, A3 Papers, A2 Papers, A1 Papers, A0 Papers, Inks,	

	Eraser, Pencil lead, Lead
Tools	Books, Internet, Handouts, Ink-pen, Pencils, set squares, Protractor, Compass, French curves, Template, Pantograph, Proportion divider, Scale rule, Pencil, sharpener, making tape, Drawing board, Knife, Scissors
Facilitation techniques	• Group discussion • Trainer guided • Matching questions • Multiple choice • Sentence completion • Short questions • Essay • Open ended questions
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment

Learning outcome 6: Perform construction drawing plans	Learning hours: 40
Indicative content	
• Dawing sheet margin and borderline • Elements of the title block • Symbols and legend • Folding drawing sheets • Construction drawing plans • Internal Perspective of drawing plans • Sections of drawing plans • Longitudinal Elements of the title block • Drawing requirements • Drawing elements • Drawing orientation • Actual location and sizes of drawing features • Specifications of materials	
Resources required for the indicative content	
Equipment	Drawing pins and clips, Compass, Drawing pencil , Drawing board, Drawing tables
Materials	Construction drawing handout, A3 Papers, A2 Papers, A1 Papers, A0 Papers, Inks, Eraser, Pencil lead, Lead
Tools	Books, Internet, Handouts, Ink-pen, Pencils, set squares, Protractor, Compass, French curves, Template, Pantograph, Proportion divider, Scale rule, Pencil, sharpener, making tape, Drawing board, Knife, Scissors

Facilitation techniques	• Group discussion • Trainer guided • Matching questions • Multiple choice • Sentence completion • Short questions • Essay • Open ended questions
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment

References:

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